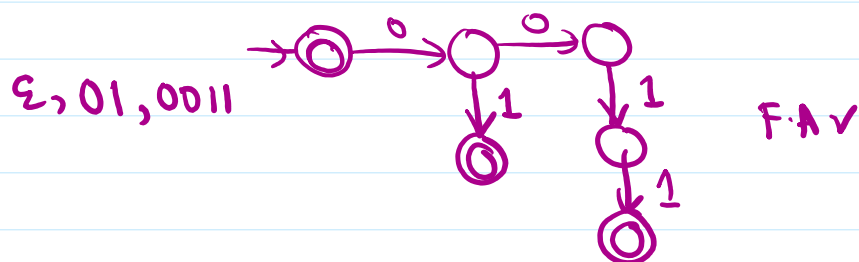


Pumping lemma

min = 0 0's
max = 2 0's

$0^n 1^n$

$n < 3$ and $n \geq 0$



PL is used to prove languages are not regular

$\{0^n 1^n \mid n \geq 0\} \quad \Sigma = \{0, 1\}$

Q. Prove this is not regular

Step ① = identify if we can make a DFA, NFA

Step ② = proof by contradiction

Step ③ = assume it's regular

Step ④ = PL will be proved ← guarantees

Step ⑤ = identify strings in this language

$0^n 1^n, 000111, 0011, \epsilon$

Step ⑥ = identify strings in terms of p

$s = 0^p 1^p$

$s = 0^{2p} 1^{2p}$

$s = 0^{5p} 1^{5p}$

Step ⑦ = break it into x.y.z

Step ⑦ = break it into x, y, z

2nd $|y| > 0$ $\swarrow$ $\searrow$ 3rd $|xy| \leq p$

OP 1^p
↓
 $|xy| \leq p$

$\frac{0}{x} \quad \frac{0}{y} \quad \frac{0^{p-2} 1^p}{z}$

2nd $|y| > 0$
length of y is greater than 0.
 $|0| = 1 > 0$ ✓ condition fulfilled

3rd $|xy| \leq p \quad |0 0| = 2 \leq p$

$\frac{0^{p-2}}{x} \quad \frac{0}{y} \quad \frac{0 1^p}{z}$

$\frac{0^1}{x} \quad \frac{0}{y} \quad \frac{0^{p-1}}{z}$

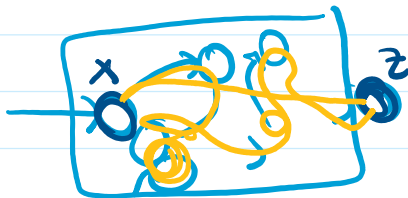
$p-1 \leq p$

Step ⑧: we are done with proving conditions
now we will pump up/down

$i=0$
 $\frac{0^{p-2}}{x} \quad \frac{0}{y} \quad \frac{0 1^p}{z}$

y ko hatado
 $i=0$ pump down

$x y^i z \quad i \geq 0 \rightarrow x y^0 z = x z$
 $= \frac{0^{p-2}}{x} \quad \frac{0}{y} \quad \frac{0 1^p}{z}$
 $= \frac{0^{p-1}}{x} \quad 1^p$



$p-1 \neq p$

$i=2$
 $x y^2 z = x y y z$
 $\nearrow 0^{p-2} \nearrow 0 \nearrow 0 1^p$
 $= \frac{0^{p-2}}{x} \quad \frac{0}{y} \quad \frac{0}{y} \quad \frac{0 1^p}{z}$
 $= \frac{0^{p-2+1+1}}{x} \quad 1^p$

Step ⑨ = that thus we have proved that P.L is not valid thus this language is not regular

$$\begin{aligned} &= \overline{0^{p^2+1+1+1}} 1^p \\ &= 0^{p+1} 1^p \end{aligned}$$

$$p+1 \neq p$$

Hence proved this string is not in the language
Because the no. of copies of 0 $\neq$ the " " 1

$$Q. \{ 0^i 1^j 0^k : i=j \text{ or } j=k \}$$

$$00110$$

$$001100$$

$$01100$$

$$S = \boxed{0^p 1^p} 0^{p+1}$$

property ✓

$$\boxed{0^{p+1}} \boxed{1^p 0^p}$$

X

$$\boxed{0^p} \boxed{1^p 0^p}$$

$|xy| \downarrow z$
 $p=p$ X

$$\boxed{0^p} \underbrace{1^p 0^{p+1}}_z$$

$$|xy| \leq p$$

$$\underbrace{0}_x \underbrace{0^{p-1}}_y \underbrace{1^p 0^{p+1}}_z$$

$$|y| > 0 \quad |0^{p-1}| = p-1 > 0 \checkmark$$

$$|xy| \leq p \quad |0 0^{p-1}| = p \leq p \checkmark$$

$$i=0 \quad xz$$

$$0 \quad 1^p 0^{p+1}$$

$$1 \neq p$$

$$\underbrace{0}_x \underbrace{0}_y \underbrace{0^{p-1} 1^p 0^{p+1}}_z$$

$$|y| > 0 = |0| = 1 > 0$$

$$|xy| \leq p = |\varepsilon 0| = 1 \leq p$$

$$i=0 \quad xz$$

$$0 \quad 0^{p-1} 1^p 0^{p+1}$$

$$p-1 \neq p$$

$$p \neq p+1$$

X

Q. NFA to DFA

part 1

Q. NFA to DFA

Power set a, b, c

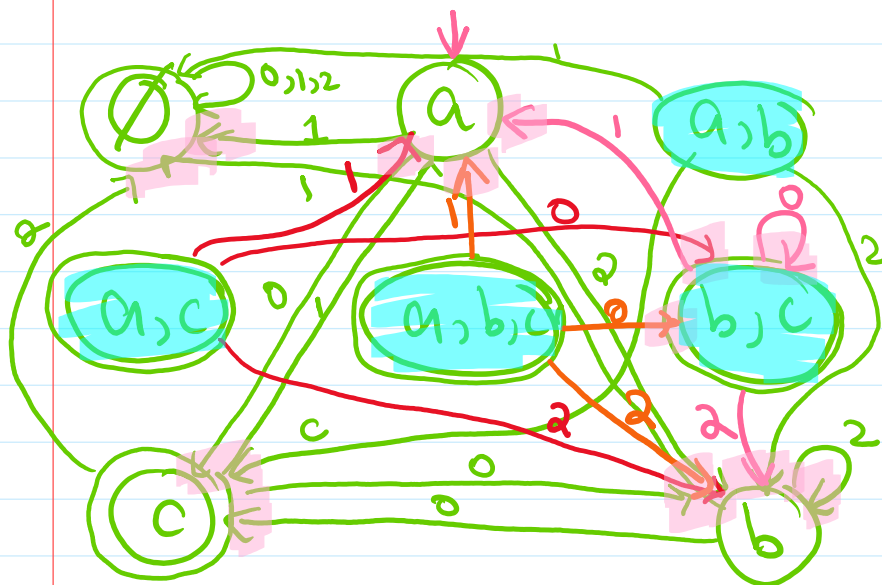
$\{ \{a\}, \{b\}, \{c\}, \{a,b\}, \{b,c\}, \{a,c\}, \{a,b,c\}, \{\emptyset\} \} = 8$

Set of all possible sets

① 1st way

3 var/set s $P.S = 2^n$
 $= 2^3 = 8$

$4 = 2^4 = 16$

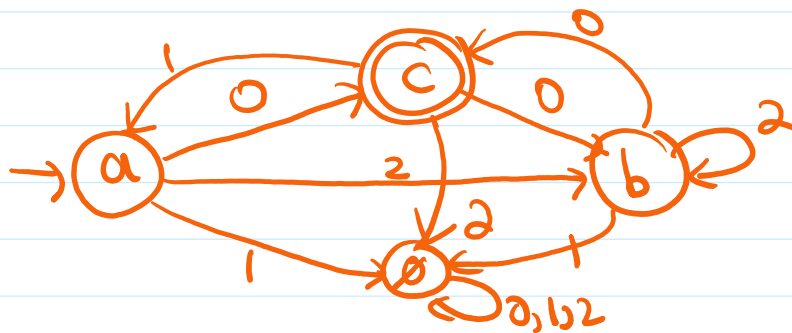


	0	1	2	
a	c	$\emptyset$	b	$\Sigma = \{0,1,2\}$ $\emptyset = \text{trap}$
b	c	$\emptyset$	b	
c	b	a	$\emptyset$	
a,b	c	$\emptyset$	b	
a,c	b,c	a	b	

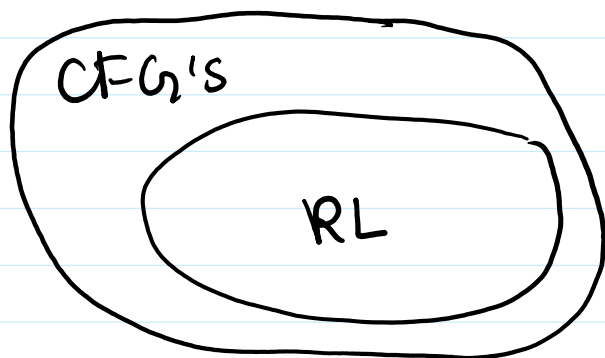
a, b	c	$\emptyset$	b
a, c	b, c	a	b
b, c	b, c	a	b
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
a, b, c	b, c	a	b

Type 2

start state	0	1	2
a	c	$\emptyset$	b
c	b	a	$\emptyset$
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
b	c	$\emptyset$	b



Minimized DFA



CFG
context free grammar



$\rightarrow A \rightarrow S A S \mid \epsilon$
 $S \rightarrow \underline{0/1}$

even

$S A S$
 $\epsilon S A S$

$0 \rightarrow 0/1$

even

$0/1$

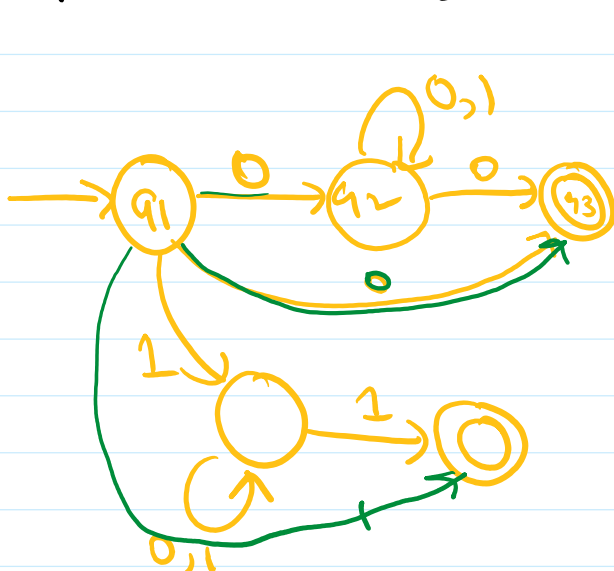
$A \rightarrow SAS$
 $\uparrow \quad \downarrow$
 $0/1 \quad SAS/\epsilon$

1001

$\begin{matrix} \text{SSASS} \\ \text{SSASS} \\ \downarrow \epsilon \end{matrix}$

This language is of all even strings

①. CFG that starts and ends with the same symbol



~~01~~, 00 010
 ||| |||

RL CFGs

$A \rightarrow 0B0 / 1C1 / 0 / 1$

$B \rightarrow BB / \epsilon / 0 / 1$
 $C \rightarrow CC / \epsilon / 0 / 1$

$\Sigma = \{0, 1\}$
 $(0\epsilon^*0) \cup (1\epsilon^*1)$
 $\cup 0 \cup 1$

A
 $0B0$
 $0BBO$
 $0BBBO$

$A \rightarrow 0B0 / 1C1 / 0 / 1$
 $B \rightarrow BB / \epsilon / 0 / 1$
 $C \rightarrow CC / \epsilon / 0 / 1$

$A \rightarrow 0B0 / 1B1 / 0 / 1 / \epsilon$
 $B \rightarrow BB / \epsilon / 0 / 1$

$A \rightarrow 0B0 \rightarrow 0BBO$
 $A \rightarrow 1B1 \rightarrow 1BB1$

$A \rightarrow 0B0 / 1B1 / C$

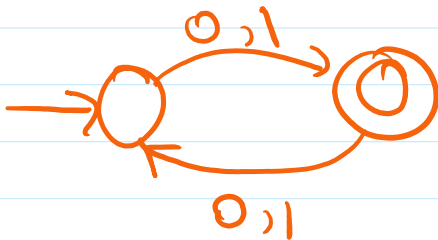
$A \rightarrow 0B0 \rightarrow 0BBO$

$A \rightarrow 0B0 \mid 1B1 \mid C$
 $B \rightarrow BB \mid \epsilon \mid C$
 $C \rightarrow 0 \mid 1$

$A \rightarrow 0B0 \rightarrow 0BBO$

Q. length is odd

$0, 1, 000, 111, 010, \dots$



DFA $(0 \cup 1)(0 \cup 1)(0 \cup 1)^*$
 $\equiv (\Sigma)^*$
 $\quad \quad \quad \epsilon$

$A \rightarrow$ [C] [C] B [C]
 $B \rightarrow$ ~~X~~ [C] ~~C~~ B C
 $C \rightarrow 0 \mid 1$

'b' len=1

$A \rightarrow '0'$

$A \rightarrow CBC$

1 C 1

1 0 1

$A \rightarrow CBC \rightarrow CAC$

$\rightarrow CCBCC$

$C C C C$

$\Sigma \Sigma \Sigma \Sigma$

$A \rightarrow CBC$
 $\rightarrow CCBCC$
 $\rightarrow CCCBCCC$

CCB

BCC

len=5